Denoising of ECG Signals Using Dual Tree Complex Wavelet Transform

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Abstract— **Electrocardiogram (ECG) signals usually corrupted by different types of noise like power line interference, baseline drift due to respiration, electromyogram interference, abrupt baseline shift and their composite noise. Denoising of noisy signal has great clinical importance for the diagnosis of cardiac abnormalities. In this paper, dual tree complex wavelet transform (DTCWT) has been used to denoise noisy ECG signals. DTCWT generates complex coefficient to obtain their real and imaginary part by using wavelet filters of dual tree. Noisy ECG signal has been simulated applying five different noise models to ECG records of MIT-BIH Arrhythmia database. The calculated SNR and MSE after denoising using DTCWT showed better performance for all types of noise than the discrete wavelet transforms (DWT). Dual tree complex wavelet transforms has performed significantly for composite of different noises.**

Keywords—ECG; noise; denoising; DTCWT; SNR

I. INTRODUCTION

ECG represents the electrical activities of heart which can offer cardiologists useful information about the functioning and the pulse of the heart. Efficient processing and analysis of ECG signals is important to detect different kind of heart activity or diseases. But contaminations of noises change the signal amplitude and frequency which may introduce serious problem to detect the actual abnormality and to take the correct decision [1]-[3].

The ECG signal may be corrupted by different types of noise like power line interference, baseline drift, EMG interference, abrupt shift noise and/or composite of them [4],[5]. Fourier transform (FT) is a very common tool for the signal processing purpose primarily but it has some limitations like it cannot process the non-stationary signals, in time domain it lacks a qualitative function, and it cannot provide time frequency resolution automatically. Another way comes to recover these problem, Short time Fourier transform (STFT) but owing to fix the window width the time frequency resolution is limited so there is no fine time frequency resolution. As the biomedical signal has low frequency and small amplitude so its small variation can cause problems and be misguided to take the correct decision.

Different denoising technique has already been applied to denoise and to extract actual information from noisy ECG signals [4]-[7]. Most of them have considered specific noise rather than group of noises. However, among different proposed techniques, performances of discrete wavelet transforms were noticeable against noisy signal [5]. DWT has several limitations like lack of shift invariance property, poor performance of distinguishing operations, aliasing, oscillations, lack of directionality. To overcome these limitations and to provide an efficient tool for denoising of the ECG signal, we plan to use dual tree complex wavelet transform technique which has elegant computational structure [8]–[11] containing shift invariance property and anti-aliasing effect specially for biomedical signal. We plan to quantify the noise susceptibility of DTCWT based denoising approach using standard ECG database and different types of noise contaminated ECG signal at different noise levels as well as intend to make comparison between performance of DTCWT and DWT approach. We also try to find perfect level of decomposition and proper level of threshold value for all types of noise reduction.

II. MATERIALS AND METHODS

A. Noise Simulation

The MIT BIH Arrhythmia database was used as source raw data [12]. The dataset has two channels recording of annotated ECG with sampling rate 360 Hz and 11 bit resolution over a 10 mV range. We have taken the records number 100, 105, 119, 200, 213 for this study. The familiar noises which contaminate the ECG signal are [5]:

- Electric signal or power line interference because of its omnipresent.
- Electromyographic (EMG) interference because of its random properties and high frequency content.
- Baseline drifts due to respiration and contains low frequency properties.
- Abrupt shift in the base line due to saturation of the amplifier in electrocardiograph.
- Finally a combination of all of the above.

For the generation of power line noise 50 Hz sinusoidal signal is added to clean ECG signal with amplitude 0.333 mV peak to peak [5]. For the baseline drift noise a very small frequency like 0.333 Hz sinusoidal signal with amplitude 1 mV

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peak to peak is added to clean ECG signal. For the EMG noisy ECG signal, a set of random number is generated consisting of values ±0.5 mV and added with clean ECG. For the abrupt shift noise a random number with value ±0.5 is generated and this number is continued to 500 samples and then another random number is created and then continued and thus so on. We have categorized the noise levels as 25%, 50%, 75% and 100% according to the amplitude levels of the noise models. Figure 1 shows the clean ECG signal and noisy signals contaminated with baseline drift, abrupt shift, power line interference, EMG noise and composite noise respectively.

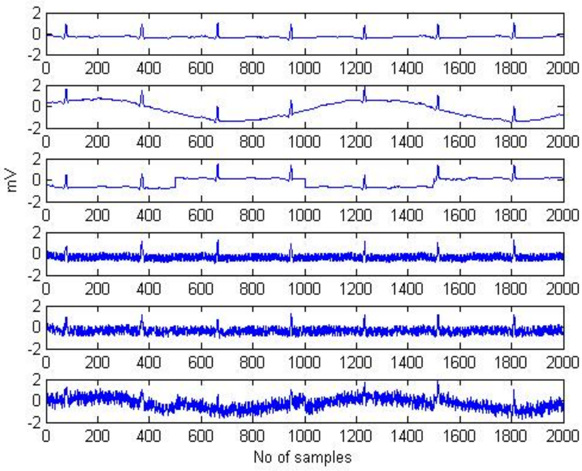


Figure 1. Clean ECG signal in the top and then noisy ECG signals contaminated with baseline drift, abrupt shift, power line interference, EMG noise and composite noise respectively.

B. *Dual Tree Complex Wavelet Transform*

The dual tree complex wavelet transform was first introduced by Kingsbury in 2001 which overcomes the main drawback of DWT for the multiresolution signal i.e. the shift variance problem of the input signal. In the shift variance problem, the small shifted of the input signal can changes the wavelet coefficients oscillation in the singularities [8]-[11]. For the lower sampling this shift variance is occurred. This can be overcomes by using undecimated form of dyadic filter tree but it requires high computational complexity and high redundancy in the output. The DTCWT with a redundancy factor 2 for 1D signal can overcome this. The DTCWT consists two trees of real filter where one for the real part and the other for the complex part of the wavelet transform. The transform is 2 times expansive than the DWT coefficient because N coefficient in DWT becomes $2N$ in DTCWT coefficients. A complex wavelet can be expressed as:

$$\Phi(t) = \Phi_h(t) + i\Phi_g(t) \qquad (1)$$

Where $\Phi_h(t)$ is the real and even, $\Phi_g(t)$ is the real and odd. $\Phi_g(t)$ is approximately Hilbert transform of $\Phi_h(t)$ [13].

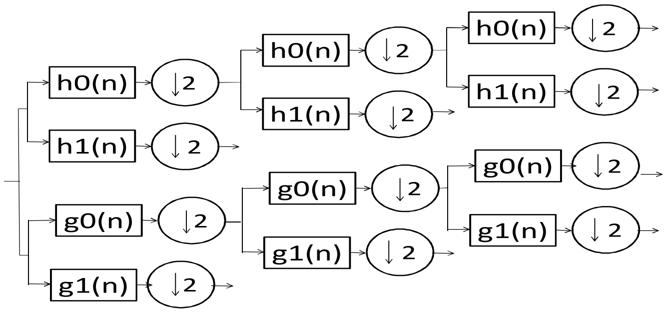
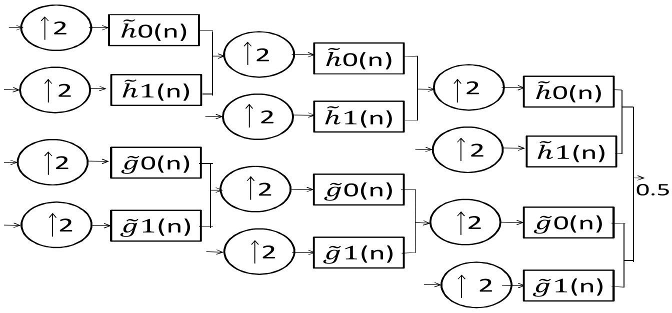


Figure 2. Analysis filter bank for DTCWT



Synthesis filter bank for DTCWT

Here the coefficients for the filters are real so complex arithmetic is not required. The inverse for the dual tree complex wavelet transform is also very easy, both real and imaginary filters are inverted separately and averaging these final reconstructed outputs can be obtained [8]. Figures 2 and 3 show the analysis and synthesis filter bank to compute the DTCWT.

C. *Denoising Using DTCWT*

For the denoising, thresholding is very important. When a signal is decomposed using wavelet transform it lefts a high frequency sub-band associated with some detail in data set. If the details are small enough, they might be omitted without affecting the main features. Additionally these small details may be noise so cutting these it actually removes noise. Similarly in low frequency sub-band if the details are small enough and those may noise can be removed by setting zero in their position. This is basic concept of thresholding. Setting all frequency sub-bands that are less than a particular value to zero and taking inverse wavelet transform to reconstruct the data set [14]. DWT noise reduction algorithm is boosted by the complex wavelet transform (CWT) which gives the substantial performance. Thresholding the coefficient of CWT is more effective rather than to the real and imaginary parts for applying magnitude nonlinearity. As the CWT provides the shift invariance property of the coefficients so the denoising algorithm also shifts invariant. Since the magnitude of CWT coefficients are strongly dependent to interscale and intrascale neighborhood so the denoising algorithm for CWT is more effective [11].

III. RESULTS ANALYSIS AND DISCUSSION

In this study, the simulations are performed using MATLAB (Mathworks Inc., USA). Different symmetric filter and quarter shift filters are used. For the denoising purpose soft thresholding is used. For the optimum threshold value, we calculate the root mean square (RMS) error between the denoised ECG signal and clean ECG signal. Different threshold values are tested iteratively and minimum RMS error containing threshold value has been selected for thresholding. The RMS error vs. threshold value curve is shown in Fig. 4.

At level 7 DTCWT have provided the best performance. Figure 5 shows the DTCWT denoised signals of 25% and 50% composite noise contaminated ECG. The base lines of noisy and filtered ECG signals are kept in level during SNR calculation for accurate comparison. Performance has been measured by calculating the signal to noise ratio (*SNR*) and mean square error (*MSE*) as follows [15]:

$$SNR = 10 * \log_{10}\left(\frac{\sum_i^N {y_i}^2}{\sum_i^N (y_i - x_i)^2}\right) \qquad (2)$$

$$MSE = \sqrt{\frac{1}{N}\sum_i^N (y_i - x_i)^2} \qquad (3)$$

Where y_i denoting the ECG signal after de-noising and x_i is the clean ECG signal.

Table I shows the *SNR* value for different types of noise at level 0%, 25 %, 50 %, 75 %, 100 %. We have considered 25% and 50% composite noise levels only as more percentages combination of all noises become severe and impractical. We have denoised the 25% and 50% composite noisy ECG signal using DWT [16] to compare with the performance of DTCWT. In DWT analysis, we have used 'Farras' wavelet with 5 levels decomposition. For effective comparison, similar thresholding approach of DTCWT has been used in DWT. Table II shows the performance comparison of DTCWT and DWT for 25% and 50% composite noisy ECG signals.

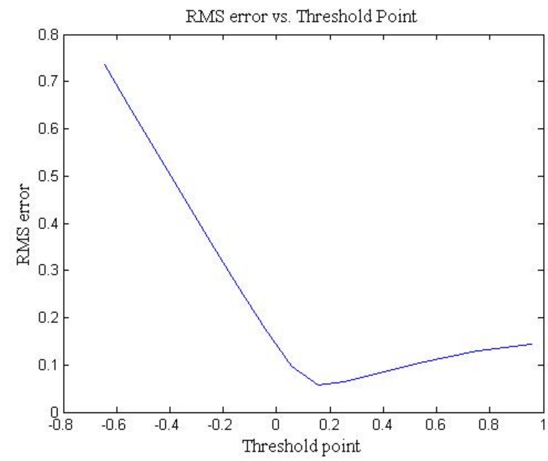


Figure 3. Threshold value selection

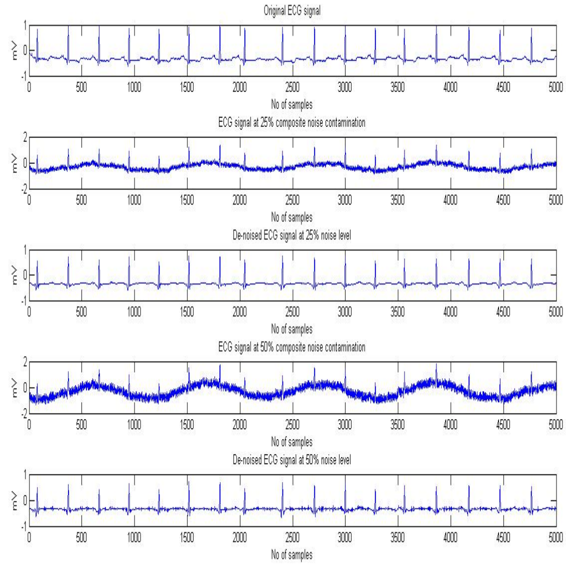


Figure 4. DTCWT denoising of composite noisy signals with 25% and 50% noise levels.

Table I shows the *SNR* value before and after denoising using DTCWT for different types of noise at level 0%, 25%, 50%, 75%, 100%. We have considered 25% and 50% composite noise levels only as more percentages combination of all noises become severe and impractical.

TABLE I. CALCULATED SNR FOR ECG SIGNALS FOR DIFFERENT TYPES OF NOISE

Noise Type	Noise Level (%)	Before de-noising SNR dB)	After de-noising SNR (dB)
No noise	0	72.56	
Power line noise	25	44.04	59.38
	50	33.69	53.12
	75	22.07	48.06
	100	16.31	44.87
Baseline drift noise	25	33.74	35.52
	50	12.27	35.44
	75	7.58	35.32
	100	5.07	35.16
EMG noise	25	39.59	52.09
	50	27.00	42.53
	75	18.82	36.09
	100	11.43	30.72
Abrupt shift noise	25	45.00	35.57
	50	33.91	35.03
	75	25.84	34.14
	100	19.53	33.74
Composite noise	25	17.86	31.49
	50	4.01	27.76

Before contaminating the noise, clean ECG signal has average *SNR* 72.56 dB but when the noise is added the *SNR* is decreased and after denoising, the SNR increase significantly but not reached to the clean ECG level.

Power line noise is very familiar noise for the ECG signal, and its denoising is comparatively easy and provides very good result at every level of noise. It lies in the decomposition level 4 of DTCWT, when 100% noise added it provide average *SNR* 44.87 dB, at 25% noise level it provide 59.38 dB.

As the characteristics of baseline drift noise are low frequency component, these can be simply eliminated by finding these low frequency and keeping them to zero. At 100% noise level 35.16 dB *SNR* obtained by DTCWT.

EMG is severe noise for ECG signal as its frequency overlaps the bandwidth of ECG. Using DTCWT techniques average *SNR* 30.72 dB has been obtained at 100% noise level. At 25 % and 50 % noise levels, 52.09 dB and 42.53 dB *SNR* were found. In the denoised ECG signal, there are some ringing signals which cannot be eliminated in case of higher level of EMG noise contamination.

For the abrupt shift noise, the method works well for 25% and 50% noise levels but discontinuity remains in shifted point in case of increased noise levels even after denoising. Average *SNR* for 100% noise level is 33.74 dB.

As composite noise is the combination of all noises mentioned above, we have considered 25% and 50% composite noise levels only since more percentages combination of all noises become severe and impractical. Though the denoising performances at higher levels have some problem but up to 50% noise level DTCWT provides good *SNR* which can also visually justified from Fig. 5.

We have denoised the 25% and 50% composite noisy ECG signal using DWT [15] to compare with the performance of DTCWT. In DWT analysis, we have used 'Farras' wavelet with 5 levels decomposition. For effective comparison, similar thresholding approach of DTCWT has been used in DWT. The comparison of *SNR* and *MSE* between the DTCWT and DWT for the 25% and 50% composite noise level is shown in Table II and Table III. It was observed that the denoising performance of DTCWT is better than the DWT. The *SNR* is about 15 dB higher as well as *MSE* is decreased. The mean square error for the DTCWT is just above 10% where as it is above 35% for the DWT.

TABLE II. COMPARISON OF *SNR* BETWEEN DWT AND DTCWT

Noise Type	Noise Level (%)	Denoising using DWT SNR (dB)	Denoising using DTCWT SNR (dB)
Composite noise	25	14.55	31.49
	50	13.91	27.76

TABLE III. COMPARISON OF *MSE* BETWEEN DWT AND DTCWT

Noise Type	Noise Level (%)	Mean square error (MSE)	
		DWT	*DTCWT*
Composite noise	0	0.3514	0.1150
	25	0.3592	0.1151
	50	0.3491	0.1319

IV. CONCLUSION

ECG signal denoising technique using dual tree complex wavelet transform is presented in this paper. The *SNR* has been calculated for different types of noises for performance analysis. The result for a given noise type can be improved if the algorithm parameters are selected specifically for that noise type. The DTCWT based method has showed improved performance for all type of noise as the *SNR* is increased after denoising the signal. DTCWT approach can be an effective way to denoise ECG as well as other physiological signals.

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